

① $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ From: Sarraj Sareef

② $\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{n+1} \cdot \frac{1}{a}$

③ $\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2\sqrt{f(x)}$

④ $\int \frac{1}{\sqrt{x^2+a^2}} dx = \ln(x + \sqrt{x^2+a^2})$

⑤ $\int \frac{1}{\sqrt{(x+a)^2 \pm b^2}} dx = \ln(x+a + \sqrt{(x+a)^2 \pm b^2})$

⑥ $\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$

⑦ $\int e^x dx = e^x + C$

⑧ $\int e^{ax} dx = \frac{e^{ax}}{a} + C$

⑨ $\int \frac{1}{1+x^2} dx = \tan^{-1}(x)$

⑩ $\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$

⑪ $\int \frac{1}{a^2+(x+b)^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x+b}{a}\right) + C$

$$(2) \int \cos x \, dx = \sin x + c$$

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$$(3) \int \sin x \, dx = -\cos x + c$$

$$(4) \int \sec x \cot x \, dx = \ln |\sec x| + c$$

$$(5) \int \sec x \tan x \, dx = \sec x + c$$

$$(6) \int \sec^2 x \, dx = \tan x + c$$

$$(7) \int \sin ax \, dx = -\frac{\cos ax}{a} + c$$

$$(8) \int \cos x \, dx = \sin x + c$$

$$(9) \int \frac{1}{1 + \sin x} \, dx \quad \text{මගේ ක්‍රමය} \rightarrow \bullet \quad t = \tan \frac{x}{2}$$

$$(10) \int u \, dv = uv - \int v \, du$$

$$(11) \int \frac{1}{x^2} \, dx = -\frac{1}{x} + c$$